

# SATQUERY AI — REMOTE SENSING ANALYSIS REPORT

ISRO Problem Statement 26167 • Mission ID: e21370a6 • Generated: 17:34:39.575Z

|                        |                                                                      |
|------------------------|----------------------------------------------------------------------|
| Natural Query:         | Compute NDVI vegetation index and canopy vigor across the spaceport. |
| Classified Task:       | SPECTRAL_INDEX                                                       |
| Calibrated Confidence: | 94% (Calibrated via Radiometric Bands)                               |
| Specialist Models:     | Spectral-Index-Analyzer-v1                                           |
| Total Latency:         | 2444.9 ms                                                            |

## GROUNDING AI INTERPRETATION

|                                                                                                                                                                                                                                                                                                                                                                                                |
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| Normalized Difference Vegetation Index (NDVI) was computed across the scene (mean NDVI: -0.0302, range: [-0.5 to 0.3297]). Quantitative analysis reveals 0.0% high-vigor healthy vegetation, 0.21% moderate grassland/shrubland canopy, and 40.99% non-vegetated surfaces (coastal waters, bare ground, and launch complexes). Overall terrain classification: Sparse / Mixed Coastal Terrain. |
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## INPUT SENSOR METADATA

| Filename              | Modality | Dimensions | Bands | GeoTIFF CRS |
|-----------------------|----------|------------|-------|-------------|
| isro_sdsc_optical.tif | OPTICAL  | 512x512    | 3     | EPSG:4326   |

## OBSERVABLE AGENTIC EXECUTION TRACE

| # | Phase / Event                                     | Observable Details                                                                                                                | Time (m) |
|---|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------|
| 1 | Query Ingestion                                   | Received natural language query: 'Compute NDVI vegetation index and canopy vigor across the spaceport.' with 1 uploaded image(s). | 0.0      |
| 2 | Input Validation                                  | Verified 1 image file(s). Format and size limits satisfied.                                                                       | 0.0      |
| 3 | Raster Metadata Extraction: isro_sdsc_optical.tif | Extracted dimensions (512x512), 3 band(s). CRS: EPSG:4326                                                                         | 102.8    |
| 4 | Modality Identification                           | Detected sensor modalities: ['OPTICAL'].                                                                                          | 0.0      |
| 5 | Query Classification → SPECTRAL_INDEX             | Single image inquiry with spectral / radiometric index mapping intent -> Routed to Spectral Index Specialist (NDVI/NDWI/NDBI).    | 4.3      |
| 6 | Model Dispatch                                    | Dispatched task to specialist: 'Spectral-Index-Analyzer-v1' on device 'cpu'.                                                      | 0.0      |
| 7 | Specialist Execution Completed                    | Inference finalized in 2330.9ms. Evidence artifacts compiled.                                                                     | 2331.0   |
| 8 | Result Fusion & Calibration                       | Derived grounded response with calibrated confidence 94% (Calibrated via Radiometric Bands).                                      | 0.0      |